

Technical report
Sparsifying l_2 regularized least squares regression
by biconjugate convex relaxation

Mamadou THIAO

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0.1 Linear least squares

$$(\text{RIDGE}^\beta) \begin{cases} \min & \|Ax - b\|_2^2 + \beta \|x\|_2^2 \\ s.t. & x \in \mathbb{R}^n, \end{cases} \quad (1)$$

where $\beta > 0$ is a parameter.

0.2 Sparsifying linear least squares

We use a l_0 regularization approach to promote sparse solutions as expressed in the following problem

$$\begin{cases} \min & \|Ax - b\|_2^2 + \rho \|x\|_0 + \beta \|x\|_2^2 \\ s.t. & x \in \mathbb{R}^n. \end{cases} \quad (2)$$

where $\rho \geq 0$ is a parameter which controls the sparsity of the solutions.

With $\delta := \sqrt{\frac{\rho}{\beta}}$, we have $\rho = \beta\delta^2$. The previous problem becomes

$$(\mathbb{P}^{\beta,\delta}) \begin{cases} \min & \|Ax - b\|_2^2 + \beta\delta^2 \|x\|_0 + \beta \|x\|_2^2 \\ s.t. & x \in \mathbb{R}^n. \end{cases} \quad (3)$$

0.3 Relaxation $(\mathbb{Q}^{\beta,\delta})$

Replacing $\|\cdot\|_0 + \frac{1}{\delta^2} \|\cdot\|_2^2$ by its tightest convex under approximation, its biconjugate,

$$\left(\|\cdot\|_0 + \frac{1}{\delta^2} \|\cdot\|_2^2 \right)^{**} (x),$$

see Le Thi et al., we obtain the following biconjugate relaxation (**BiConjugate Relaxation**, **BCR**)

$$(\mathbb{Q}^{\beta,\delta}) \left\{ \begin{array}{ll} \min & F(x) := \|Ax - b\|_2^2 + \beta\delta^2 \left(\|\cdot\|_0 + \frac{1}{\delta^2} \|\cdot\|_2^2 \right)^{**} (x) \\ \text{s.t.} & x \in \mathbb{R}^n. \end{array} \right.$$

$(\mathbb{Q}^{\beta,\delta})$ is a convex under approximation of $(\mathbb{P}^{\beta,\delta})$ and $(\mathbb{Q}^{\beta,\delta})$ solves $(\mathbb{P}^{\beta,\delta})$ when its solution satisfies particular global optimality conditions (see Le Thi et al.).

$$\left(\|\cdot\|_0 + \frac{1}{\delta^2} \|\cdot\|_2^2 \right)^{**} (x) = \sum_i^n \left(|\cdot|_0 + \frac{1}{\delta^2} |\cdot|_2^2 \right)^{**} (x_i),$$

$$\left(|\cdot|_0 + \frac{1}{\delta^2} |\cdot|_2^2 \right)^{**} (x_i) = \frac{1}{\delta^2} x_i^2 - \frac{1}{\delta^2} \left[(\delta - |x_i|)^+ \right]^2 + 1 = \begin{cases} \frac{1}{\delta^2} x_i^2 + 1 & \text{if } |x_i| \geq \delta. \\ 2\frac{1}{\delta} |x_i| & \text{otherwise.} \end{cases}$$

0.3.1 Optimality conditions for $(\mathbb{Q}^{\beta,\delta})$

The following form for the optimality conditions are very useful for the theoretical study of the links between the biconjugate relaxation BCR, Ridge, Lasso and ElasticNet.

Optimality condition for $(\mathbb{Q}^{\beta,\delta})$

The optimality condition for the convex relaxation $(\mathbb{Q}^{\beta,\delta})$ is given by, Le Thi et al.

$$-2A^T(Ax - b) \in \beta\delta^2 \partial \left(\|\cdot\|_0 + \frac{1}{\delta^2} \|\cdot\|_2^2 \right)^{**} (x).$$

As

$$\partial \left(|\cdot|_0 + \frac{1}{\delta^2} |\cdot|_2^2 \right)^{**} (x_i) = \begin{cases} \left\{ 2\frac{1}{\delta^2} x_i \right\} & \text{if } |x_i| \geq \delta, \\ \left\{ 2\frac{1}{\delta} \text{sign}(x_i) \right\} & \text{if } 0 < |x_i| \leq \delta, \\ \left[-2\frac{1}{\delta}, 2\frac{1}{\delta} \right] & \text{if } x_i = 0, \end{cases}$$

the condition becomes

$$\left(\mathbb{Q}_{opt}^{\beta, \delta}\right) [A^T(Ax - b)]_i \in \begin{cases} \{-\beta x_i\} & \text{if } |x_i| \geq \delta \\ \{-\delta\beta \text{ sign}(x_i)\} & \text{if } 0 < |x_i| \leq \delta \\ [-\delta\beta, \delta\beta] & \text{if } x_i = 0. \end{cases}$$

or

$$[A^T(Ax - b) + \beta x]_i \in \begin{cases} \{0\} & \text{if } |x_i| \geq \delta \\ \{\beta x_i - \delta\beta \text{ sign}(x_i)\} & \text{if } 0 < |x_i| \leq \delta \\ [-\delta\beta, \delta\beta] & \text{if } x_i = 0. \end{cases}$$

Proposition 0.3.1.1 $x = 0$ will be an optimal solution for $(\mathbb{Q}^{\beta, \delta})$ if and only if $\|A^T b\|_\infty \leq \delta\beta$.

Proof: Direct use of $\left(\mathbb{Q}_{opt}^{\beta, \delta}\right)$. □

TO DO: study for the unicity or not of the solution!!!

Proposition 0.3.1.2 If x is a solution for $(\mathbb{Q}^{\beta, \delta})$ and $\delta < \min_{i \in \text{supp}(x)} |x_i|$, then x is a solution for $(\mathbb{Q}^{\beta, \delta'})$, for all $\delta', \delta \leq \delta' \leq \min_{i \in \text{supp}(x)} |x_i|$.

Proof: Direct use of $\left(\mathbb{Q}_{opt}^{\beta, \delta}\right)$. □

0.3.2 Theoretical links with Ridge and Lasso

Theoretical links with Ridge

Recall

$$(\text{RIDGE}^\beta) \begin{cases} \min & \|Ax - b\|_2^2 + \beta \|x\|_2^2 \\ \text{s.t.} & x \in \mathbb{R}^n. \end{cases} \quad (4)$$

Ridge optimality condition $(2A^T(Ax - b) + 2\beta x = 0)$ can be written as

$$\left(\text{RIDGE}_{opt}^\beta\right) [A^T(Ax - b)]_i \in \{-\beta x_i\}.$$

Proposition 0.3.2.1 A RIDGE is always a $(\mathbb{Q}^{\beta, \delta})$.

More precisely, assume A has no null column,

1. If x is a solution for (RIDGE^β) , then x is a solution for $(\mathbb{Q}^{\beta, \delta})$ for all $\delta \leq \min_i |x_i|$.
2. If x is a solution for $(\mathbb{Q}^{\beta, \delta})$ and $\delta \leq \min_i |x_i|$, then x is a solution for (RIDGE^β) .

Proof: Direct use of $\left(\text{RIDGE}_{opt}^\beta\right)$ and $\left(\mathbb{Q}_{opt}^{\beta,\delta}\right)$. $\square$

Theoretical links with Lasso

$$(\text{LASSO}^\gamma) \begin{cases} \min & \|Ax - b\|_2^2 + \gamma \|x\|_1 \\ \text{s.t.} & x \in \mathbb{R}^n. \end{cases} \quad (5)$$

Lasso optimality condition can be written as

$$\left(\text{LASSO}_{opt}^\gamma\right) [A^T(Ax - b)]_i \in \begin{cases} \left\{-\frac{\gamma}{2} \text{sign}(x_i)\right\} & \text{if } x_i \neq 0 \\ \left[-\frac{\gamma}{2}, \frac{\gamma}{2}\right] & \text{if } x_i = 0. \end{cases}$$

Proposition 0.3.2.2 *A LASSO is always a $(\mathbb{Q}^{\beta,\delta})$.*

More precisely,

1. *If x is a solution for (LASSO^γ) ,
then x is a solution for $(\mathbb{Q}_{2\delta}^{\gamma,\delta})$ for all $\delta \geq \max_{i \in \text{supp}(x)} |x_i|$.*
2. *If x is a solution for $(\mathbb{Q}^{\beta,\delta})$ and $\delta \geq \max_{i \in \text{supp}(x)} |x_i|$,
then x is a solution for $(\text{LASSO}^{2\delta\beta})$.*

Proof: Direct use of $\left(\text{LASSO}_{opt}^\gamma\right)$ and $\left(\mathbb{Q}_{opt}^{\beta,\delta}\right)$. $\square$

From Ridge to Lasso. How navigate $(\mathbb{Q}^{\beta,\delta})$

Consider (RIDGE^β) , for a $\beta > 0$. (Assume A has no null column)
Define

$$\delta_{\text{RIDGE}^\beta} := \min_{i \in \text{supp}(x^{\text{RIDGE}^\beta})} \left| x_i^{\text{RIDGE}^\beta} \right|.$$

Let us analyze δ s values.

We already have x^{RIDGE^β} solution of $(\mathbb{Q}^{\beta,\delta})$ for all $\delta \leq \delta_{\text{RIDGE}^\beta}$.

We will consider $\delta_{\text{RIDGE}^\beta} > \delta$.

From inequality $F(x^{\mathbb{Q}^{\beta,\delta}}) \leq F(0)$, we deduce

$$\left| x_i^{\mathbb{Q}^{\beta,\delta}} \right| \leq \left\| x^{\mathbb{Q}^{\beta,\delta}} \right\|_1 \leq \frac{1}{2\delta\beta} F(0), \forall i.$$

So it suffices to constrain $\frac{1}{2\delta\beta} F(0) \leq \delta$ to ensure $\left| x_i^{\mathbb{Q}^{\beta,\delta}} \right| \leq \delta$, for all i .

So $x^{\mathbb{Q}^{\beta,\delta}}$ is a solution of $(\text{LASSO}^{2\delta\beta})$, for all $\delta > \max \left\{ \sqrt{\frac{1}{2\beta}F(0)}, \delta_{\text{RIDGE}^\beta} \right\}$.

Define

$$\delta_{\text{RIDGE}^\beta, \text{LASSO}} := \inf \left\{ \delta > \delta_{\text{RIDGE}^\beta} : x^{\mathbb{Q}^{\beta,\delta}} \text{ is a solution of } (\text{LASSO}^{2\delta\beta}) \right\}.$$

We also have necessarily

$$\rho_{\text{RIDGE}^\beta, \text{LASSO}} \leq \max \left\{ \sqrt{\frac{1}{2\beta}F(0)}, \delta_{\text{RIDGE}^\beta} \right\}.$$

Proposition 0.3.2.3 1. When δ moves from $\delta_{\text{RIDGE}^\beta}$ to $\delta_{\text{RIDGE}^\beta, \text{LASSO}}$, $(\mathbb{Q}^{\beta,\delta})$ moves from *RIDGE* to *LASSO*.

- before $\delta_{\text{RIDGE}^\beta}$, $(\mathbb{Q}^{\beta,\delta})$ is a *RIDGE*,
- after $\delta_{\text{RIDGE}^\beta, \text{LASSO}}$, $(\mathbb{Q}^{\beta,\delta})$ is a *LASSO*.

2. When δ moves strictly between $\delta_{\text{RIDGE}^\beta}$ and $\delta_{\text{RIDGE}^\beta, \text{LASSO}}$, $(\mathbb{Q}^{\beta,\delta})$ cannot be a *RIDGE* nor a *LASSO*.

The components of $x^{\mathbb{Q}^{\beta,\delta}}$ satisfy $\left(\mathbb{Q}^{\beta,\delta}_{\text{opt}} \right)$

$$\left[A^T (Ax^{\mathbb{Q}^{\beta,\delta}} - b) \right]_i \in \begin{cases} \left\{ -\beta x_i^{\mathbb{Q}^{\beta,\delta}} \right\} & \text{if } \left| x_i^{\mathbb{Q}^{\beta,\delta}} \right| \geq \delta \\ \left\{ -\delta \beta \text{sign}(x_i^{\mathbb{Q}^{\beta,\delta}}) \right\} & \text{if } 0 < \left| x_i^{\mathbb{Q}^{\beta,\delta}} \right| \leq \delta \\ [-\delta\beta, \delta\beta] & \text{if } x_i^{\mathbb{Q}^{\beta,\delta}} = 0. \end{cases}$$

The second point of the previous proposition highlights the mixing role (on the components of x) played by $(\mathbb{Q}^{\beta,\delta})$. The first line of the optimality condition is for *RIDGE*, with β , and the followings are for *LASSO*, with $\gamma := 2\delta\beta$. The value $\delta = \frac{\gamma}{2\beta}$ defines a trade-off line on the coordinates value.

0.3.3 Mixing roles: $(\mathbb{Q}^{\beta,\delta})$ vs $(\text{ELASTICNET}^{\beta',\gamma})$

Theoretical links with ElasticNet

$$(\text{ELASTICNET}^{\beta',\gamma}) \begin{cases} \min & \|Ax - b\|_2^2 + \beta' \|x\|_2^2 + \gamma \|x\|_1 \\ \text{s.t.} & x \in \mathbb{R}^n. \end{cases} \quad (6)$$

ElasticNet optimality condition can be written as

$$\left(\text{ELASTICNET}^{\beta',\gamma}_{\text{opt}} \right) \left[2A^T (Ax - b) \right]_i \in \begin{cases} \{ -2\beta' x_i - \gamma \text{sign}(x_i) \} & \text{if } x_i \neq 0 \\ [-\gamma, \gamma] & \text{if } x_i = 0. \end{cases}$$

Proposition 0.3.3.1 x is a common solution for $(\text{ELASTICNET}^{\beta',\gamma})$ and $(\mathbb{Q}^{\beta,\delta})$ if and only if x is a solution of $(\text{ELASTICNET}^{\beta',\gamma})$ or $(\mathbb{Q}^{\beta,\delta})$ and one of the following conditions holds:

1. there exists C_1 , such that

$$|x_i| = C_1, \forall i, x_i \neq 0,$$

$$C_1 \geq \delta, \beta = \beta' + \frac{\gamma}{2C_1}.$$

2. there exists C_2 , such that

$$|x_i| = C_2, \forall i, x_i \neq 0,$$

$$C_2 \leq \delta, \delta\beta = \frac{\gamma}{2} + \beta'C_2.$$

3. there exist C_1 and C_2 , $C_2 \leq C_1$, such that

$$|x_i| = C_1 \text{ or } C_2, \forall i, x_i \neq 0,$$

$$\delta = \frac{C_1(\gamma + 2\beta'C_2)}{\gamma + 2\beta'C_1}, \delta\beta = \frac{\gamma}{2} + \beta'C_2.$$

Proof: Use $\left(\text{ELASTICNET}_{opt}^{\beta', \gamma}\right)$, $\left(\mathbb{Q}_{opt}^{\beta, \delta}\right)$ and compare solutions. $\square$

0.4 Solving $(\mathbb{Q}^{\beta, \delta})$

Let us denote by $x^\beta(\delta)$ a solution for $(\mathbb{Q}^{\beta, \delta})$.

Proposition 0.4.0.1 *Let $\beta > 0$ and fixed.*

$x^\beta(\delta)$ is a piecewise linear function (TO DO why not mapping? think of it!!!) with respect to $\delta \geq 0$.

Proof: ... $\square$

0.4.1 Computing a new solution from a known one

Here we will consider that we have $(\mathbb{Q}^{\beta, \underline{\delta}})$, $\underline{\delta} > 0$ and a corresponding solution $\underline{x} := x^\beta(\underline{\delta})$.

Let us define the following partition of the indices i , $i = 1, \dots, n$.

$$\begin{aligned}
\mathbb{K}_0 &:= \{i : |\underline{x}_i| > \underline{\delta}\}, \\
\mathbb{K}_1 &:= \{i : |\underline{x}_i| = \underline{\delta}, \text{Type 1}\}, \\
\mathbb{K}_2 &:= \{i : |\underline{x}_i| = \underline{\delta}, \text{Type 2}\}, \\
\mathbb{K}_3 &:= \{i : 0 < |\underline{x}_i| < \underline{\delta}\}, \\
\mathbb{K}_4 &:= \{i : \underline{x}_i = 0, |[A^T(A\underline{x} - b)]_i| = \underline{\delta}\beta, \text{Type 1}\}, \\
\mathbb{K}_5 &:= \{i : \underline{x}_i = 0, |[A^T(A\underline{x} - b)]_i| = \underline{\delta}\beta, \text{Type 2}\}, \\
\mathbb{K}_6 &:= \{i : \underline{x}_i = 0, 0 < |[A^T(A\underline{x} - b)]_i| < \underline{\delta}\beta\}, \\
\mathbb{K}_7 &:= \{i : \underline{x}_i = 0, [A^T(A\underline{x} - b)]_i = 0\}, \\
\underline{M} &:= \text{diag}(\underline{\theta}), \\
\underline{\theta}_i &:= \text{sign}(\underline{x}_i) \begin{cases} +1 & \text{if } \underline{x}_i > 0, \\ -1 & \text{if } \underline{x}_i < 0, \\ -\text{sign}([A^T(A\underline{x} - b)]_i) & \text{if } [A^T(A\underline{x} - b)]_i \neq 0, \\ +1 & \text{otherwise.} \end{cases}
\end{aligned}$$

The following polyhedra contains $u := |\underline{x}|$ and $\delta := \underline{\delta}$, so is nonempty.

$$\begin{array}{llllll}
(A^T \underline{A} \underline{M})_i u + \beta \underline{\theta}_i \underline{u}_i & & & = (A^T b)_i & i \in \mathbb{K}_0 \\
(A^T \underline{A} \underline{M})_i u + \beta \underline{\theta}_i \underline{u}_i & & & = (A^T b)_i & i \in \mathbb{K}_1 \\
(A^T \underline{A} \underline{M})_i u + \beta \underline{\theta}_i \underline{u}_i & + \beta \underline{\theta}_i t_i & & = (A^T b)_i & i \in \mathbb{K}_2 \\
(A^T \underline{A} \underline{M})_i u + \beta \underline{\theta}_i \underline{u}_i & + \beta \underline{\theta}_i t_i & & = (A^T b)_i & i \in \mathbb{K}_3 \\
(A^T \underline{A} \underline{M})_i u + \beta \underline{\theta}_i \underline{u}_i & + \beta \underline{\theta}_i t_i & & = (A^T b)_i & i \in \mathbb{K}_4 \\
(A^T \underline{A} \underline{M})_i u & + \beta \underline{\theta}_i s_i & & = (A^T b)_i & i \in \mathbb{K}_5 \\
(A^T \underline{A} \underline{M})_i u & + \beta \underline{\theta}_i s_i & & = (A^T b)_i & i \in \mathbb{K}_6 \\
(A^T \underline{A} \underline{M})_i u & - \beta \underline{\theta}_i s_i & + \beta \underline{\theta}_i w_i & = (A^T b)_i & i \in \mathbb{K}_7 \\
u_i & & - t_i & = \delta & i \in \mathbb{K}_0 \\
u_i & & - t_i & = \delta & i \in \mathbb{K}_1 \\
u_i & & + t_i & = \delta & i \in \mathbb{K}_2 \\
u_i & & + t_i & = \delta & i \in \mathbb{K}_3 \\
u_i & & + t_i & = \delta & i \in \mathbb{K}_4 \\
& s_i & + w_i & = \delta & i \in \mathbb{K}_5 \\
& s_i & + w_i & = \delta & i \in \mathbb{K}_6 \\
& s_i & + w_i & = \delta & i \in \mathbb{K}_7
\end{array}$$

$$u \geq 0 \quad \delta \geq 0 \quad s \geq 0 \quad t \geq 0 \quad w \geq 0 \quad .$$

For $x := \underline{M}u$,

$$[A^T(Ax - b) + \beta x]_i = \begin{cases} 0 & 0 & i \in \mathbb{K}_0 \\ 0 & 0 & i \in \mathbb{K}_1 \\ -\beta \underline{\theta}_i t_i & \beta x - \delta \beta \text{sign}(x_i) & i \in \mathbb{K}_2 \\ -\beta \underline{\theta}_i t_i & \beta x - \delta \beta \text{sign}(x_i) & i \in \mathbb{K}_3 \\ -\beta \underline{\theta}_i t_i & \beta x - \delta \beta \text{sign}(x_i) & i \in \mathbb{K}_4 \\ -\beta \underline{\theta}_i s_i & -(\delta \beta - w_i \beta) \text{sign}(x_i) & i \in \mathbb{K}_5 \\ -\beta \underline{\theta}_i s_i & -(\delta \beta - w_i \beta) \text{sign}(x_i) & i \in \mathbb{K}_6 \\ \beta \underline{\theta}_i s_i - \beta \underline{\theta}_i w_i & -(-\delta \beta + 2w_i \beta) \text{sign}(x_i) & i \in \mathbb{K}_7 \end{cases}$$

Proposition 0.4.1.1 $x := \underline{M}u$ satisfies $(\mathbb{Q}^{\beta, \delta})$ optimality conditions for δ sufficiently near $\underline{\delta}$.

Proof: ... □

How to choose δ ?

Our goal is to maximize $|\delta - \underline{\delta}|$ while keeping the polyhedral constraints valid.

Two cases $\delta := \underline{\delta} - \gamma$ or $\delta := \underline{\delta} + \gamma$, $\gamma \geq 0$. The first one will be used when starting from zero solution. The second one will be from Ridge solution.

Define τ ,

$$\tau := \begin{cases} +1 & \text{when starting from zero solution,} \\ -1 & \text{when starting from Ridge solution.} \end{cases}$$

So $\delta := \underline{\delta} - \tau\gamma$, in all cases.

We will maximize $|\delta - \underline{\delta}| = |-\tau\gamma| = \gamma$, over the previous polyhedra with green and red ones as basis, in all cases.

max γ

$(A^T \underline{AM})_i u + \beta \underline{\theta}_i u_i$				$= (A^T b)_i$	$i \in \mathbb{K}_0$
$(A^T \underline{AM})_i u + \beta \underline{\theta}_i u_i$				$= (A^T b)_i$	$i \in \mathbb{K}_1$
$(A^T \underline{AM})_i u + \beta \underline{\theta}_i u_i$		$+\beta \underline{\theta}_i t_i$		$= (A^T b)_i$	$i \in \mathbb{K}_2$
$(A^T \underline{AM})_i u + \beta \underline{\theta}_i u_i$		$+\beta \underline{\theta}_i t_i$		$= (A^T b)_i$	$i \in \mathbb{K}_3$
$(A^T \underline{AM})_i u + \beta \underline{\theta}_i u_i$		$+\beta \underline{\theta}_i t_i$		$= (A^T b)_i$	$i \in \mathbb{K}_4$
$(A^T \underline{AM})_i u$	$+\beta \underline{\theta}_i s_i$			$= (A^T b)_i$	$i \in \mathbb{K}_5$
$(A^T \underline{AM})_i u$	$+\beta \underline{\theta}_i s_i$			$= (A^T b)_i$	$i \in \mathbb{K}_6$
$(A^T \underline{AM})_i u$	$-\beta \underline{\theta}_i s_i$	$+\beta \underline{\theta}_i w_i$		$= (A^T b)_i$	$i \in \mathbb{K}_7$
u_i	$+\tau\gamma$		$-\underline{t}_i$	$= \underline{\delta}$	$i \in \mathbb{K}_0$
u_i	$+\tau\gamma$		$-\underline{t}_i$	$= \underline{\delta}$	$i \in \mathbb{K}_1$
u_i	$+\tau\gamma$		$+\underline{t}_i$	$= \underline{\delta}$	$i \in \mathbb{K}_2$
u_i	$+\tau\gamma$		$+\underline{t}_i$	$= \underline{\delta}$	$i \in \mathbb{K}_3$
u_i	$+\tau\gamma$		$+\underline{t}_i$	$= \underline{\delta}$	$i \in \mathbb{K}_4$
	$\tau\gamma$	$+s_i$		$+\underline{w}_i$	$= \underline{\delta}$
	$\tau\gamma$	$+s_i$		$+\underline{w}_i$	$= \underline{\delta}$
	$\tau\gamma$	$+s_i$		$+\underline{w}_i$	$= \underline{\delta}$
$u \geq 0$	$\gamma \geq 0$	$s \geq 0$	$t \geq 0$	$w \geq 0$	.

After Gauss elimination, for each row with 0 value on the right hand side and a nonnegative one on the γ column, do as follow

$$\begin{aligned} \text{if } i \in \mathbb{K}_1 &\Rightarrow \mathbb{K}_1 := \mathbb{K}_1 \setminus \{i\}, \mathbb{K}_2 := \mathbb{K}_2 \cup \{i\} \\ \text{if } i \in \mathbb{K}_2 &\Rightarrow \mathbb{K}_2 := \mathbb{K}_2 \setminus \{i\}, \mathbb{K}_1 := \mathbb{K}_1 \cup \{i\} \\ \text{if } i \in \mathbb{K}_4 &\Rightarrow \mathbb{K}_4 := \mathbb{K}_4 \setminus \{i\}, \mathbb{K}_5 := \mathbb{K}_5 \cup \{i\} \\ \text{if } i \in \mathbb{K}_5 &\Rightarrow \mathbb{K}_5 := \mathbb{K}_5 \setminus \{i\}, \mathbb{K}_4 := \mathbb{K}_4 \cup \{i\} \end{aligned}$$

and perform another Gauss elimination.

Proposition 0.4.1.2 *At most 2 eliminations are performed to reach optimality.*

Proof: ... □

Proposition 0.4.1.3 *Let $\beta > 0$, fixed.*

- $\underline{x} := 0$ is a solution for $(\mathbb{Q}^{\beta, \delta})$,

$$\text{for all } \delta \geq \delta^0 := \frac{1}{\beta} \|A^T b\|_{\infty}.$$

- $\underline{x} := x^{\text{Ridge}^{\beta}}$ is a solution for $(\mathbb{Q}^{\beta, \delta})$,

$$\text{for all } \delta \leq \delta^{\text{Ridge}^{\beta}} := \min_{x_i^{\text{Ridge}^{\beta}} \neq 0} |x_i^{\text{Ridge}^{\beta}}|.$$

- From $\underline{\delta} := \delta^0$ and $\underline{x} := 0$, repeating the process until Ridge solution is attained gives us the full path and solutions for $(\mathbb{Q}^{\beta, \delta})$, $\delta \geq 0$ varying. Ridge solution is attained whenever $\underline{\delta} = \min_{i=1, \dots, n} |\underline{x}_i|$. (Assuming matrix A has no zero column)
- From $\underline{\delta} := \delta^{\text{Ridge}^{\beta}}$ and $\underline{x} := x^{\text{Ridge}^{\beta}}$, repeating the process until zero solution is attained gives us the full path and solutions for $(\mathbb{Q}^{\beta, \delta})$, $\delta \geq 0$ varying. Zero solution is attained whenever $\underline{\delta} = \delta^0$.

Proof: ... □

	$\mathbb{K}_0$	$\mathbb{K}_1$	$\mathbb{K}_2$	$\mathbb{K}_3$	$\mathbb{K}_4$	γ^1	b^1
$\mathbb{K}_0$	?	?	?	?	?	γ^1	b^1
$\mathbb{K}_1$	?	?	?	?	?	γ^1	b^1
$\mathbb{K}_2$	?	?	$-\beta M_{K_2}$	?	?	$\gamma^1 - \beta M_{K_2} \gamma^2$	$b^1 - \beta M_{K_2} b^2$
$\mathbb{K}_3$	?	?	?	$-\beta M_{K_3}$	?	$\gamma^1 - \beta M_{K_3} \gamma^2$	$b^1 - \beta M_{K_3} b^2$
$\mathbb{K}_4$	?	?	?	?	$-\beta M_{K_4}$	$\gamma^1 - \beta M_{K_4} \gamma^2$	$b^1 - \beta M_{K_4} b^2$
$\mathbb{K}_5$	?	?	?	?	?	γ^1	b^1
$\mathbb{K}_6$	?	?	?	?	?	γ^1	b^1
$\mathbb{K}_7$	?	?	?	?	?	$\gamma^1 - \beta M_{K_3} \gamma^3$	$b^1 - \beta M_{K_3} b^3$
						γ^2	b^2
$\mathbb{K}_0$	I_{K_0}	0	0	0	0	γ^2	b^2
$\mathbb{K}_1$	0	I_{K_1}	0	0	0	γ^2	b^2
$\mathbb{K}_2$	0	0	I_{K_2}	0	0	γ^2	b^2
$\mathbb{K}_3$	0	0	0	I_{K_3}	0	γ^2	b^2
$\mathbb{K}_4$	0	0	0	0	I_{K_4}	γ^2	b^2
						γ^3	b^3
$\mathbb{K}_5$	0	0	0	0	0	γ^3	b^3
$\mathbb{K}_6$	0	0	0	0	0	γ^3	b^3
$\mathbb{K}_7$	0	0	0	0	0	γ^3	b^3

We denote by γ'' and b'' the columns corresponding to γ and right hand side b after Gauss elimination.

Define a diagonal matrix D^1 by

$$D_{ii}^1 := \begin{cases} -1 & \text{if } i \in \mathbb{K}_0 \cup \mathbb{K}_1, \\ +1 & \text{otherwise.} \end{cases}$$

	$\mathbb{K}_0$	$\mathbb{K}_1$	$\mathbb{K}_2$	$\mathbb{K}_3$	$\mathbb{K}_4$	$\gamma^1 \text{'''}$	$b^1 \text{'''}$
$\mathbb{K}_0$	I_{K_0}	0	0	0	0	$\gamma^1 \text{''}$	$b^1 \text{''}$
$\mathbb{K}_1$	0	I_{K_1}	0	0	0	$\gamma^1 \text{''}$	$b^1 \text{''}$
$\mathbb{K}_2$	0	0	I_{K_2}	0	0	$\gamma^1 \text{''}$	$b^1 \text{''}$
$\mathbb{K}_3$	0	0	0	I_{K_3}	0	$\gamma^1 \text{''}$	$b^1 \text{''}$
$\mathbb{K}_4$	0	0	0	0	I_{K_4}	$\gamma^1 \text{''}$	$b^1 \text{''}$
$\mathbb{K}_5$	0	0	0	0	0	$\frac{1}{\beta} M_{K_5} \gamma^1 \text{''}$	$\frac{1}{\beta} M_{K_5} b^1 \text{''}$
$\mathbb{K}_6$	0	0	0	0	0	$\frac{1}{\beta} M_{K_6} \gamma^1 \text{''}$	$\frac{1}{\beta} M_{K_6} b^1 \text{''}$
$\mathbb{K}_7$	0	0	0	0	0	$-\frac{1}{2\beta} M_{K_7} \gamma^1 \text{''}$	$-\frac{1}{2\beta} M_{K_7} b^1 \text{''}$
						$\gamma^2 \text{'''}$	$b^2 \text{'''}$
$\mathbb{K}_0$	0	0	0	0	0	$D^1(\gamma^2 \text{''} - \gamma^1 \text{''})$	$D^1(b^2 \text{''} - b^1 \text{''})$
$\mathbb{K}_1$	0	0	0	0	0	$D^1(\gamma^2 \text{''} - \gamma^1 \text{''})$	$D^1(b^2 \text{''} - b^1 \text{''})$
$\mathbb{K}_2$	0	0	0	0	0	$D^1(\gamma^2 \text{''} - \gamma^1 \text{''})$	$D^1(b^2 \text{''} - b^1 \text{''})$
$\mathbb{K}_3$	0	0	0	0	0	$D^1(\gamma^2 \text{''} - \gamma^1 \text{''})$	$D^1(b^2 \text{''} - b^1 \text{''})$
$\mathbb{K}_4$	0	0	0	0	0	$D^1(\gamma^2 \text{''} - \gamma^1 \text{''})$	$D^1(b^2 \text{''} - b^1 \text{''})$
						$\gamma^3 \text{'''}$	$b^3 \text{'''}$
$\mathbb{K}_5$	0	0	0	0	0	$\gamma^3 \text{''} - \frac{1}{\beta} M_{K_5} \gamma^1 \text{''}$	$b^3 \text{''} - \frac{1}{\beta} M_{K_5} b^1 \text{''}$
$\mathbb{K}_6$	0	0	0	0	0	$\gamma^3 \text{''} - \frac{1}{\beta} M_{K_6} \gamma^1 \text{''}$	$b^3 \text{''} - \frac{1}{\beta} M_{K_6} b^1 \text{''}$
$\mathbb{K}_7$	0	0	0	0	0	$\gamma^3 \text{''} + \frac{1}{2\beta} M_{K_7} \gamma^1 \text{''}$	$b^3 \text{''} + \frac{1}{2\beta} M_{K_7} b^1 \text{''}$

0.5 Sparse Principal Component

Consider a covariance matrix Q .

Writing $Q = A^T A$ and $b := A\alpha$, we can see that

$$\|Ax - b\|_2^2 = (x - \alpha)^T Q(x - \alpha),$$

$$A^T(Ax - b) = Q(x - \alpha)$$

and

$$A^T b = Q\alpha.$$

0.6 Conclusion